Teaching Philosophy Statement

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My primary goal as an instructor is to make sure every student feels welcome and respected, and knows they belong in the classroom. Students arrive with very different experiences of mathematics. I want to help them build confidence and find satisfaction in understanding something that once seemed difficult, and I am especially committed to supporting students from underrepresented and underprivileged communities. I share the convictions behind Federico Ardila-Mantilla's axioms for mathematics education: that mathematical talent is distributed equally across groups of people, and that every student deserves meaningful mathematical experiences and to be treated with dignity and respect.[1]

My own interest in mathematics took time to develop. As a child, I attended weekend coaching classes taught by a physics teacher who traveled more than twenty miles to reach us, a journey of over two hours in Indian city traffic. I struggled with mathematics and physics then, but I still remember the experiments he showed us, and as I solved more problems and began asking my own questions, I fell in love with mathematics. Those experiences remind me how much a teacher can do by making room for curiosity, and they are why I volunteer with math circles and with science and mathematics camps for schoolchildren.

Making Mistakes Is Part of Learning. When a student is unsure of an answer or makes a mistake, my first concern is to help them feel at ease. I remind the class that an incomplete idea is still worth discussing, and I openly acknowledge my own mistakes. When someone is stuck, I offer small hints and ask questions that help them find a next step. I want students to see revising an argument as part of doing mathematics, and to feel comfortable sharing their reasoning before they are certain it is correct. One student wrote, "He treats all the students respectfully and replies to our queries quickly."

Motivating New Ideas. To help students into an unfamiliar idea, I use geometric pictures and applications before a formal definition. I use a stack of pancakes, for instance, to show why the volume of a cylinder is its base area times its height, and then use the same picture to motivate the double integral. As instructor of record for MTH 314, Matrix Algebra with Applications, at Michigan State University in Summer 2024, I used geometric transformations of the plane to motivate eigenvalues and eigenvectors. One student wrote that "we learned what these concepts actually represented in linear algebra rather than just the routine calculations." This captures a central goal of my teaching: students should be able to explain what their calculations mean.

Active Learning and Group Work. After introducing a concept and working an example, I give students time to try a related worksheet problem before I present a solution. I circulate, offer hints, help students find a partner, and watch where their reasoning stalls. When several groups struggle with the same idea, I bring the class together at the board for a brief, targeted explanation, which uses class time efficiently and shows students that their questions are shared. A classroom observer described the result as a "collaborative and productive classroom environment."

In Spring 2025, I taught approximately sixty students as instructor of record for MTH 103A, College Algebra I, the first half of a two-semester stretch sequence for students who benefit from additional time and structured in-class instruction. This format let us explore core algebraic concepts deeply while reinforcing prerequisites as they arose. I prepared my own lecture notes and worksheets and worked with two undergraduate learning assistants. The course used standards-based grading on a pass/fail basis, with further opportunities to demonstrate mastery, so an unsuccessful attempt became a reason to keep working rather than a final verdict, which in my experience reduced students' stress.

Teaching Online and Caring for the Whole Student. I taught MTH 124, Survey of Calculus, fully online in Summer 2023 as a flipped classroom: students watched videos I recorded before class, so our live sessions could be spent working through problems together; optional self-introduction videos helped classmates find study partners. In MTH 314, also online, I did not wait for struggling students to come to me; I wrote to

[1] Federico Ardila-Mantilla, *Todos Cuentan: Cultivating Diversity in Combinatorics*, Notices of the AMS **63** (2016), no. 10, 1164–1170.

them and met with them individually on Zoom. My students have lives and pressures outside my course, so I offer flexible deadlines in difficult circumstances, space exams so that students have a fair chance to prepare, and, when MTH 103A met at 8 a.m. through a Michigan winter, offered a Zoom option for icy mornings.

Supporting the Transition to Proof. As a recitation leader for MTH 317H, Honors Linear Algebra, I teach proof techniques that students need but may not yet know, with worksheets that ask why a concept is useful or needed. In Fall 2026, I redesigned these worksheets around problem posing: students change hypotheses, formulate their own questions, and exchange problems with classmates. Students keep their worksheets, which include definitions and notes to study from, and hand in only a short exit ticket. Since the change, participation has increased and students have responded positively. The worksheets are at `s-adithya.github.io/teaching/mth317h-fs26.html`, and in Spring 2027, as a mathematics education mini-project for the Certification in College Teaching, I plan to conduct a thorough study of how problem posing affects participation and the explanations students give for their reasoning.

Teaching with AI. With the rise of generative AI, it is especially important to design activities that reveal students' thinking. Where course policies permit, I would invite students to critique an AI-generated solution, identify unsupported steps, and explain how to correct them. Students could also use AI to suggest examples, while checking its claims against course materials. I would make expectations for acknowledging AI assistance explicit. So that access never limits participation, I would use AI tools that the institution provides free to all students, or open-source alternatives, and students who prefer not to use AI would have alternative pathways to earn the same credit. Written reasoning and discussion would remain central to assessment.

Mathematics Education Background. I am working toward the Certification in College Teaching at Michigan State University, including coursework in Teaching College Mathematics. Discussions and reflections with classmates from several disciplines have encouraged me to offer students more choice in how they demonstrate understanding, such as a video, artwork, or physical model with a written explanation of the mathematics.

Mentoring and Leadership. In Summer 2026, I mentored two undergraduates in the Michigan State University Summer Topology Program. Explaining chain complexes to a student who had not studied abstract algebra, I learned to build a path into advanced ideas from familiar vector spaces and linear maps. As a teaching assistant for graduate algebra, I helped first-year Ph.D. students prepare for their qualifying examination. I have spent six semesters as a tutor and supervisor in the Mathematics Learning Center, and as Lead Teaching Assistant in 2024–25, I observed and gave feedback to eight teaching assistants and sixteen undergraduate learning assistants.

Future Teaching and Mentoring. Working closely with students, both in the classroom and through undergraduate research, is central to how I hope to teach and mentor. I would welcome opportunities to contribute across the undergraduate curriculum, including general education courses and courses for majors, drawing on my experience in college algebra, calculus, linear algebra, and proof-based instruction, and to develop advanced offerings connected to my interests in algebra and topology. Having earned a certificate from the Erdős Institute Data Science Boot Camp, I would be glad to teach statistics, data science, differential equations, and real and complex analysis, and I am expanding my knowledge of machine learning and deep learning. I am especially interested in curriculum development and mathematics pedagogy, and would welcome collaborating with colleagues on new courses and teaching methods. To involve more students in research, I would create entry points through guided reading and exploratory projects that help them move toward questions of their own.

Across these settings, from building confidence in College Algebra and guiding proof-writing recitations to supporting graduate algebra students and mentoring undergraduate topology projects, I keep returning to what that physics teacher showed me: students learn best when someone makes room for their curiosity and believes they can understand. I will keep listening to student feedback and adapting, and I hope students leave my courses more confident asking questions, exploring ideas, and doing mathematics together. My teaching portfolio is available at `s-adithya.github.io/teaching/portfolio.html`.